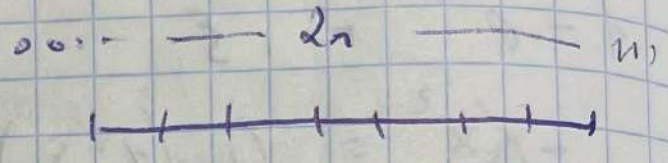
Лекция 4



$C = \{c_1, \dots, c_m\} \in 2^n$

2^k

$$\left\{\begin{matrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{matrix}\right\} \xrightarrow[a \in C]{} \begin{matrix} c_1 + ? = \\ \vdots \\ c_m + ? = \end{matrix} \left\{\begin{matrix} c_i \\ \vdots \\ c_j \end{matrix}\right\} \qquad \left\{\begin{matrix} c_1 \\ \vdots \\ c_m \end{matrix}\right\} \xrightarrow[b \notin C]{} \left\{\begin{matrix} c_m + a \\ c_1 + b \\ \vdots \\ c_m + b \end{matrix}\right\}$$

$+ b \neq 0$

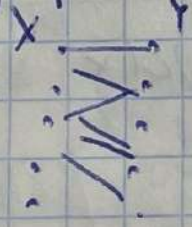
$c_i + a = c_j + b$

$c_i + c_j = a + b$

$\in C$

Декодир по макс. правдоподоб.

ДСК



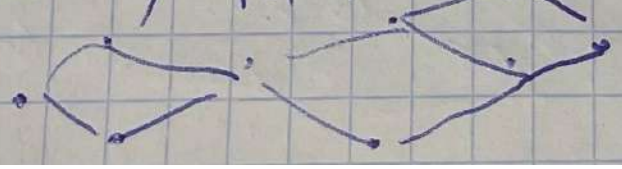
$\{p(y|x),\ x \in X,\ y \in Y\}$

$\mathbb{C} = \{c_i,\ i = 1, \dots, M\} \subseteq X^n$

$\bar{y} = (y_1 \dots y_n)$ ф. правдоподоб $\to$ max

$p(\bar{y} | \bar{c}_i) \quad i = 1, \dots, M$

$$p(\bar{y} | \bar{x}) = \prod_{i=1}^{n} p(y_i | x_i)$$

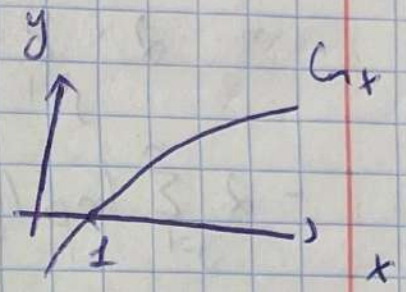


Логарифм. ф.п.

$$\Lambda_m(\bar{y}) = \ln p(\bar{y}/\bar{c}_m) = \sum_{i=1}^{n} \ln p(y_i/c_{m_i})$$

$$= \sum_{i=1}^{n} L_m(y_i)$$

$$L(y) = \ln \frac{p(y|1)}{p(y|0)}$$



Выбор надежности

$$L(\bar{y}) = (L(y_1), \dots, L(y_n))$$

$$\xi_m = \{\xi_{m1}, \dots, \xi_{mn}\}$$

$$\bar{\xi}_{m_i} = 2 \cdot c_{m_i} - 1$$

$$c_{m_i} = 0 \to -1$$

$$c_{m_i} = 1 \to 1$$

$$(\bar{\xi}_m, L(\bar{y}))$$

Т. Для ДСК декодир по мах. правдоп сводится к поиску код. слова, для кот. макс. $L_m(\bar{y}) = (\bar{\xi}_m, L(\bar{y}))$

$\bar{c}_m$ и $\bar{c}_{m'}$, $m \neq m'$ $\quad \Lambda_m(\bar{y}) \geq \Lambda_{m'}(\bar{y}) \Rightarrow$

$\Rightarrow L_m(\bar{y}) \geq L_{m'}(\bar{y})$

$c_m = (0 \dots)$ 1 2 3 ...

I_{00} $c_{m_i} = 0$

$c_{m_i} = 0$ $c_{m'} = (0 \dots)$

$I_{10} = \{i\}$ $\quad I_{11} =$ $\quad I_{10} = k$

$$L_m(\bar{y}) - L_{m'}(\bar{y}) = \sum_{j \in I_{10}} ((+1) - (-1)) \cdot L(y_j) +$$

$$+ \sum_{j \in I_{01}} ((-1) - (+1)) \cdot L(y_j) = 2 \cdot \sum_{j \in I_{10}} (y_j) - 2 \sum_{j \in I_{01}} L(y_j) =$$

$$L(y_j) = \ln \frac{p(y_j|1)}{p(y_j|0)} \quad = 2 \cdot \sum_{j \in I_{10}} \ln \frac{p(y_j/\bar{c}_{m_j})}{p(y_j/\bar{c}_{m'_j})} -$$

$$-2\sum_{j\in I_{01}} \ln \frac{p(y_i|\overline{c_{m'}}_j)}{p(y_i|\overline{c_m}_j)} =$$

$$= 2\cdot\sum_{j=1}^{n} \ln \frac{p(y|c_{m'j})}{p(y|c_{mj})} = 2\sum_{j=1}^{n} \ln p(y|c_{m'j}) -$$

$$-2\sum_{j=1}^{n} \ln p(y|c_{mj}) = 2\left(\Lambda_m(\bar y) - \Lambda_{m'}(\bar y)\right)$$

$$\Lambda_m(\bar y) - \Lambda_{m'}(\bar y) \sim L_m(\bar y) - L_{m'}(\bar y)$$

ДСК $\quad p_0 \qquad L(y) = \ln \frac{p(y|1)}{p(y|\varnothing)}$

$$L(y) = \begin{cases} \ln \frac{1-p_0}{p_0} = \delta, & y = 1 \\ \ln \frac{p_0}{1-p_0}, & y = \varnothing \end{cases}$$

$$L_m(\bar y) = (\xi_m, L(\bar y)) = \quad y_i = c_{mi} \to \delta$$

$$= \sum_{i=1}^{n} (2c_{mi}-1)\, L(y_i) = \quad \begin{matrix} y_i = 1 & c_{mi} = 1 \to 1 \\ y_i = 0 & c_{mi} = 0 \to -1 \end{matrix}$$

$$= d(\bar y, \bar c_m)\cdot(-\delta) + (n - d(\bar y, \bar c_m))\cdot\delta =$$

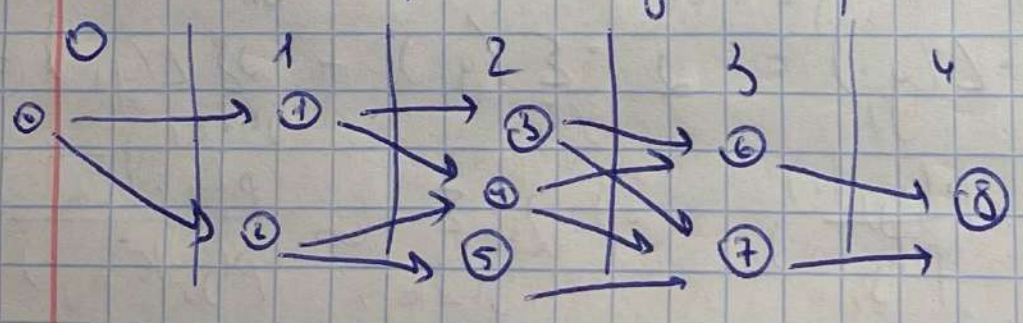
$$= \delta\left(-d(\bar y, \bar c_m) + n - d(\bar y, \bar c_m)\right) =$$

$$= \delta\left(n - 2d(\bar y, \bar c_m)\right) \to \max$$

$p_0 < \frac{1}{2}$, $\delta > 0$ $\qquad$ min $\Rightarrow$ Декод по мин р $X_{\text{ЭМ}}$.

$=$ Декод по макс. правд. в ДСК

Поиск кратч. пути в решёт. Алг. Витерби



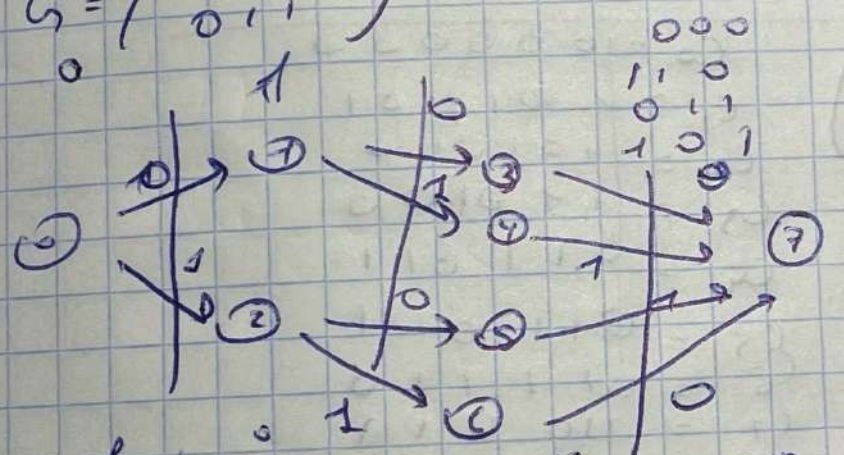
← ориг

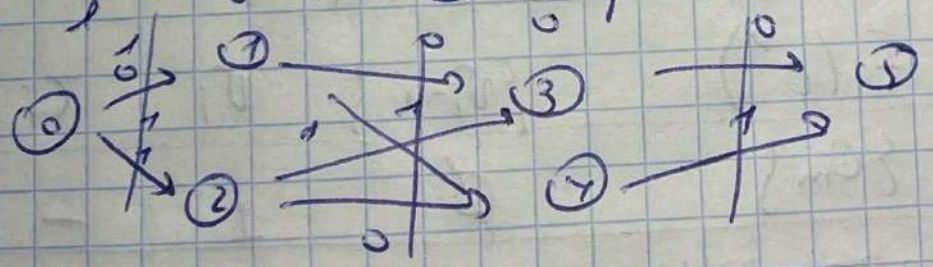
свойства:

1) Верш. разб. на непересек. мн-ва (ярусы)
2) Нулев и послед. ор. содерж. 1 узел
3) Решетк — направл. граф
4) Ребра ~~...~~ при матрице
5) Длина пути = Σ длин ребер в этот путь

Линейное блоковое код

$G = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ (3,2) с. прав на четность





Чем меньше узлов, тем эффект в декод

Сложность решетки → пропускн

$\tau = (\tau_0, \dots, \tau_n)$ (1, 2, 2, 1)

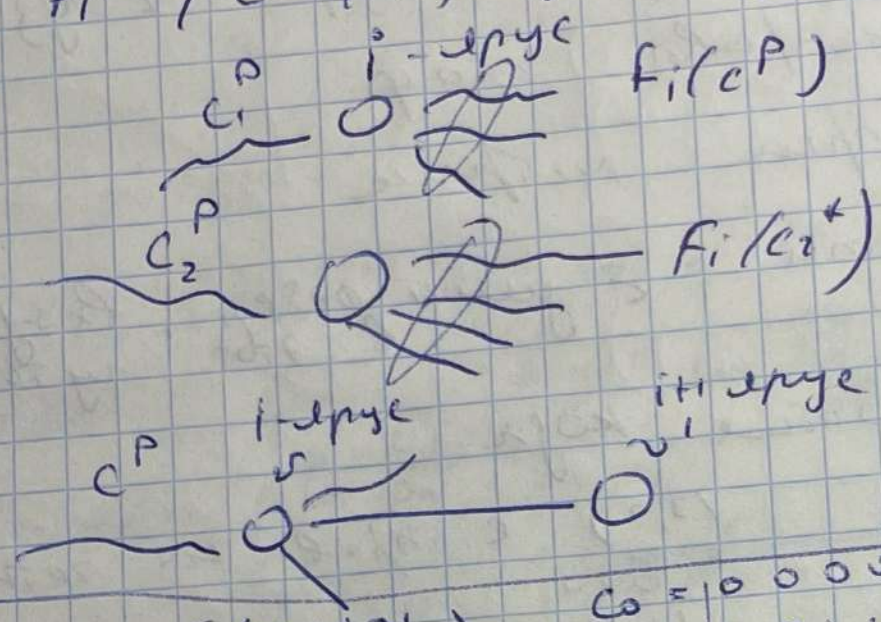
$c = \{c_m, m = \overline{1, M}\}$ длин. $\underline{n}$

$\bar{c}_m = (c_{m1}, c_{m2}, \dots, c_{mn})$

Разобьем кодов. слово на 2 части

$\bar{c}_m = (c_m^p, c_m^f)$ p = past, f = future

$F_i = \{ c^f : (c^p, c^f) \in C$ que tient c^p



$$G = \begin{pmatrix} 1\,0\,1\,1\,0\,1 \\ 1\,0\,1\,0\,1\,0 \\ 1\,1\,0\,1\,0\,0 \end{pmatrix}$$

$c_0 = 0\,0\,0\,0\,0\,0$
$c_1 = 1\,0\,1\,1\,0\,1$
$c_2 = 1\,0\,1\,0\,1\,0$
$c_3 = 1\,1\,0\,1\,0\,0$
$c_4 = 0\,1\,1\,0\,1\,1$
$c_5 = 0\,1\,1\,0\,0\,1$
$c_6 = 0\,1\,1\,1\,1\,0$
$c_7 = 1\,1\,0\,0\,1\,1$

7 ésycol

i	$\{c^p\}$	$F_i(c^p)$	$\{v_{i-1}\}$	v_i	c_i
0	$\{\emptyset\}$	$\{c_m\}$	$\emptyset$	1	–
1	0	0000, 0111 1001, 1110	1	1, 2	0
1	1	0110 1, 0101 0 1010 1, 1001 1	1	2, 3	1
2	00	0000, 0111	1^2	1^4	0
	01	1001, 1110	1^2	2^5	1
	10	1101, 1010	2^3	3^2	0
	11	0100, 0011	2^3	4^7	1